

Sentinel Intrusion Detection

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Real-Time Brute-Force Defense for High-Traffic University Portals

The Problem:

During result release week, university portals face aggressive bot brute-force attacks guessing student passwords.

Impact:

High volume of fraudulent login attempts exhausts server resources and degrades service for legitimate students.

Current Limitation:

Distributed intrusion prevention tools (like CrowdSec) introduce polling latency between bouncers and LAPI.

Sentinel's Mission:

A focused, single-process intrusion prevention system enforcing exact, zero-delay ban expiration at request-time.

Solution & Core Flow

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Streamlined Detection-to-Enforcement Pipeline

Demo Portal (POST /login):

Authenticates against seeded credentials and emits one structured log line per attempt.

Exact Log Format:

{timestamp} | ip={ip} | username={username} | result=success|fail appended to portal_auth.log.

Real-time Ingestion:

Async Log Watcher tails log file and feeds the Log Parser to emit structured event records.

Sliding-Window Detection:

Attack Detector tracks per-IP failure frequency within sliding time window (WINDOW_SECONDS=60).

Instant Ban Creation:

When failures reach BAN_THRESHOLD=10, writes Alert and Decision records directly into SQLite.

Architecture & Technology Stack

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Python 3.11+ | FastAPI | SQLite | pytest

Unified Process Monolith:

No distributed agent/LAPI orchestration; all components communicate in-process with minimal overhead.

Blocker ASGI Middleware:

Positioned directly in front of Demo Portal; evaluates client IP on every incoming request.

Request-Time Enforcement:

Evaluates 'WHERE value = :ip AND until > :now' via composite index (value, type, until).

403 Forbidden Response:

Banned requests are rejected immediately with JSON payload containing exact ban expiration ('until').

Decoupled Expiry Sweeper:

Background task periodically cleans expired database records without affecting live access.

The Three Killer Tests

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Rigorously Verified via Automated Pytest & Manual Demo Scripts

Killer Test 1 (Brute-Force Detection):

10 failed logins from 198.51.100.10 within 60s trigger ban; attempt #11 receives HTTP 403.

Killer Test 2 (Innocent Bystander Isolation):

Attacker (198.51.100.10) is banned while innocent student (203.0.113.50) logs in with HTTP 200 OK.

Killer Test 3 (Exact Expiry):

Short ban duration (e.g. 2s) proves ban lifts immediately at $t > \text{until}$ via per-request check.

Dual Execution Modes:

Every test is verifiable both programmatically via pytest and interactively via demo curl scripts.

Improvements & Differentiator

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Clear Engineering Advancements Over Existing Solutions

Gap Fix (Exact Request-Time Expiry):

CrowdSec bouncers poll periodically, causing bans to linger past expiration. Sentinel's middleware eliminates polling delay.

Decoupled Maintenance:

Blocker middleware check makes unblocking millisecond-exact; the Sweeper is purely for database garbage collection.

Differentiator (AI Threat Explainer):

Synthesizes attack patterns into plain-English incident narratives for IT administrators.

Resilient & Optional:

System operates with 100% functionality without an AI key; zero secrets or external dependencies required.